Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Force

Meaning in my words How would you explain force to a friend?	**How can you tell?** When your hand pushes a cart, what pushes and what receives the push?
Example or sketch Which force arrow would you draw while a hand pushes a cart right?	**Non-example and why** The hand lets go and the cart keeps rolling. Is the hand still pushing it?

Net force

Meaning in my words How would you explain net force to a friend?	**How can you tell?** Two people push one cart in opposite directions. Why do the arrow directions matter?
Example or sketch One person pushes 6 N right and another 4 N left. What single arrow replaces both?	**Non-example and why** A student adds those pushes to get 10 N right. Which push did they count the wrong way?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Newton's 1st law

Meaning in my words What does Newton's first law say in your own words?	**How can you tell?** A cart is already moving and its forces balance. What stays the same?
Example or sketch Draw a cart at three one-second intervals as it coasts right with zero net force.	**Non-example and why** A classmate says a rolling cart needs a forward net force every second. What would that force actually change?

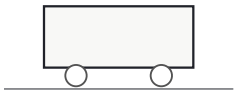
Inertia

Meaning in my words How would you explain inertia to a friend?	**How can you tell?** An empty cart and a loaded cart start at rest. The same net force pushes each for one second. Which gains less speed? Why?
Example or sketch With balanced forces, where will a rolling cart be one second later and one second after that?	**Non-example and why** Someone points to the hand pushing the cart and says, "That push is inertia." What would you tell them?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw a moving cart with balanced force arrows in orange and its separate velocity arrow in purple.



My marks mean

One camera. Which window?

09

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use timed sightings to make a prediction, then check it.

CHOOSE ONE JOB TOGETHER

TRACKING ENGINEER ☐

Radio a camera window. Point to the sightings that support your call.

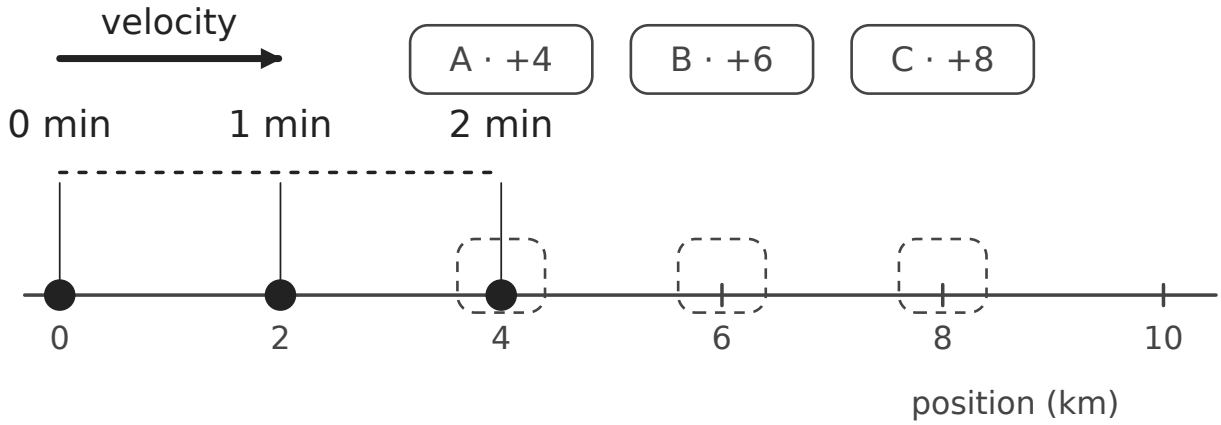
SCIENCE BROADCASTER ☐

Give a 15-second update. Use your cart drawing to explain why XJ can keep moving without an engine.

Both partners point to a clue. Ask: What has to stay the same for our call to work?

OBSERVATION DESK

One picture at minute 3. Choose A, B or C.



01 OUR FIRST CALL

Where should the camera look? Why?
Write your radio call or on-air update.

02 OUR REVISED CALL

Wait for your teacher. Keep your first call.

New time: _____ min Position: _____ km

What will you tell them now?
Use the new sighting in your update.

On the graphic: add a dashed circle for your forecast. After the reveal, add a star for the new sighting. Color the velocity arrow **PURPLE**. Use **PENCIL** for data marks.

Switch speakers. Explain your correction while your partner points to the new mark.

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks